

Project Requirement Document

Project: CampusConnect — College Event & Community Platform

This document describes the initial business and technical requirements for building CampusConnect, a web platform that allows college students to discover events, register for events, communicate with organizers, and manage their participation.

1. Business Objective

Build a centralized platform for college students and event organizers. Students should be able to discover upcoming events, view event details, register, and receive notifications. Organizers should be able to create and manage events and monitor registrations.

2. Target Users

- Students: discover events, register, manage registrations, and receive notifications.
- Event Organizers: create events, update event information, and view registered participants.
- Administrator: manage users, events, reports, and platform-level settings.

3. Functional Requirements

ID	Requirement
FR-01	Users must be able to register and log in using email and password.
FR-02	Users should be able to update their profile information.
FR-03	Students should be able to browse and search upcoming events.
FR-04	Each event should contain a title, description, date, time, location, organizer, and maximum capacity.
FR-05	Students should be able to register for an event if seats are available.
FR-06	Students should be able to cancel their event registration.
FR-07	Organizers should be able to create, edit, and cancel events.
FR-08	Organizers should be able to view the list of registered students.
FR-09	The system should send email notifications after successful registration.
FR-10	Administrators should be able to deactivate users and remove inappropriate events.

4. Non-Functional Requirements

- The platform should support at least 5,000 registered users.
- Typical API responses should complete within 2 seconds under normal load.

- Passwords must never be stored in plain text.
- Role-based access control must restrict organizer and administrator functionality.
- The system should maintain logs for important administrative actions.
- The application should be deployable using Docker.

5. Technology Preferences

- Frontend: React
- Backend: Python with FastAPI
- Database: PostgreSQL
- Cache/temporary data: Redis
- Authentication: JWT or an equivalent secure authentication mechanism
- Deployment: Docker-based deployment

6. Constraints

- Initial development timeline: 10 weeks.
- The first release should focus on core event discovery and registration.
- The system should be designed so that mobile application support can be added later.
- The first release should not include online payment processing.

7. Future Scope

- Mobile application for Android and iOS.
- Event recommendations based on student interests.
- QR-based attendance tracking.
- Analytics dashboard for organizers.
- Integration with college calendars.